MILITARY SPECIFICATION

ARMING DEVICE MARK 6 MOD 0 (TORPEDO PRACTICE)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 This specification governs the manufacture, assembly and preparation for delivery of Arming Device Mark 6 Mod 0 (Torpedo Practice) and the methods of inspection and tests upon which product acceptance will be based. Arming Device Mark 6 Mod 0 (Torpedo Practice) is an assembly adapted for receiving two electrorepsonsive actuators which are shunted when the arming device is in a disarmed position and electrically connected in a firing circuit when the arming device is in an armed position.

2. APPLICABLE DOCUMENTS

2.1 The following documents; of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

Federal Test Method Standard No. 406	Plastics: Method of Testing

STANDARDS

MILITRAY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

FSC 1355

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DRAWINGS

BUREAU OF NAVAL WEAPONS

LD 291742	Arming Device Mark 6 Mod 0 Torpedo Practice" Assembly (Empty)
LD 291935	Preserved (Packaging) Condition Arming Device Mark 6 Mod 0 (Loaded)
LD 291936	Shipping (Packing) Condition for 24 Arming Devices Mark 6 Mod 0 (Loaded)
Dwg 883562	Contact Mount Assembly
Dwg 883581	Retaining Ring
Dwg 959425	Spring Washer
Dwg 959486	Safety Spring
Dwg 1378167	Arming Device Mark 6 Mod O Torpedo Practice Assembly (Empty)
Dwg 1387994	Safety Spring Releasing Tool

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.

3.1.1 Arming Device Mark 6 Mod 0 (Torpedo Practice) shall be manufactured in accordance with LD 291742 and all documents listed thereon.

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3.1.2 Unless otherwise specified by the procuring agency a pre-production sample shall be required from the contractor as provided in 4.1.2.

3.2 Inert assembly. An assembly of Arming Device Mark 6 Mod O (Torpedo Practice) shall be comprised of the parts listed on LD 291742.

3.2.1 Cleanliness. Immediately prior to the assembly of the arming device, each component shall be free of dirt, dust, flux, oil, oxides, sulfides, or any other foreign matter that may impair assembly or subsequent functioning of the apparatus.

3.2.2 Degree of cure of plastic components. The plastic upper and lower contact mounts and plastic shutter shall be thoroughly cured and shall have an acetone extraction value of 5 percent maximum when tested in accordance with 4.5.3.1.

3.2.3 Soldering assembly. When soldering the contact fingers sub-assembly, the solder shall be confined to an area over and above the riveted portion of the pin and in no instance shall it be allowed to flow between the contact fingers in a manner that will prevent their acting as independent cantilever springs. During assembly of the contact fingers, the resistor leads shall be made mechanically secure prior to soldering. The contact fingers shall be tinned and soldered together at their interfaces as the resistor leads are soldered. Solder shall not flow between the contact fingers. It is recommended that a suitable fixture be provided for properly orienting the resistors and the contact fingers of the contact fingers assembly during the performance of the soldering operation. Immediately after soldering, all flux shall be cleaned from the soldered and surrounding surfaces. An examination of the assembly shall indicate the absence of loose solder or such condition that could conceivably result in a short or circuit failure.

3.2.4 Fracture strength. The plastic shutter, when assembled within the contact mount and inserted with the contact mount into the housing without the safety spring, Dwg 959486, shall withstand a torque of 30 inch-pounds applied to the free end of the shutter shaft in either direction without fracture of either the shutter or mount or a loosening of the shutter shaft. The spring, Dwg 959486, shall not be “inserted except at final assembly.

3.2.5 Housing, shutter, spring and contact mount assembly. The shutter and contact mount assembly, as assembled within the housing without spring, shall fall by gravity free of the housing as the housing is held, open end downward, with its axis substantially vertical. The shutter and contact mount assembly, as assembled within the housing with

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spring, shall be free to be drawn from the housing as the safety spring releasing tool, Dwg 1387994 is inserted through the 0.125 hole in the side of the housing and the spring is sufficiently compressed.

3.2.6 Operating torque. The shutter, including dummy detonators, when assembled with the contact mount assembly but without the safety spring, shall rotate from the unarmed stop position to the armed stop position and from the armed stop position to the unarmed stop position with a torque of 4 plus or minus 2 inch pounds.

3.2.7 Switch securement. After the switch has been secured by the bracket to the housing, as called for on Dwg 1378167, the switch shall resist a 3 pound pull, applied to the leads of the switch, without shifting position beneath the bracket or coming free of the housing.

3.2.8 Switch operation. The switch, when secured by the bracket to the housing, shall close as a force of 4 pounds is applied directly against the switch actuating face to show an electrical resistance between the switch leads of less than 0.1 ohm.

3.2.9 Armed position resistance. When dummy detonators, which have been internally shorted to show a resistance of less than 0.05 ohm between the prongs, are inserted in the shutter and rotated to the armed position, the resistance between the terminal pins of the contact mount assembly shall be not greater than 0.1 ohm.

3.2.10 Contact mount resistance, With detonators removed, the resistance between the two terminals of the contact mount assembly shall be not less than 15.4 ohms nor greater than 17.4 ohms.

3.2.11 Unarmed position resistance. When individual leads, whose resistances are not more than 0.03 ohm each, are attached to the prongs of a dummy detonator having open circuit characteristics and the detonator is inserted in the shutter and rotated to the disarmed position, the resistance between the detonator leads shall be not greater than 0.1 ohm.

3.2.12 Dielectric strength. On the contact mount assembly, Dwg 883562, the insulation resistance between either terminal pin and the rivets of the shunt fingers shall be not less than 10 megohms. When the contact mount assembly and shutter are assembled in the housing, without detonators and without the safety spring, the insulation resistance between either terminal pin and the shutter shaft, and between either terminal pin and the housing, shall be not less than 10 megohms.

3.2.13 Durability requirements. Arming Device Mark 6 Mod 0 (Torpedo Practice) shall be capable of meeting acceptably all the requirements of this specification after being subjected to the vibration, shock, temperature and storage tests of 4.6.1, 4.6.2, 4.6.3, and 4.6.4, respectively.

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However, the contractor will not be held responsible for failure to meet the durability requirements specified except when such failure is attributable to nonconformance with the drawings.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection, shall be in accordance with the provisions set forth in MIL-STD-105.

4.1.2 Preproduction sample. Unless otherwise specified for each contract or order, a representative sample of five arming devices shall be manufactured using the methods and procedures proposed for the production lot. These shall be submitted to the Government inspector for preproduction tests at an activity designated in the contract or by the Bureau of Naval Weapons for purposes of determining compliance with the requirements of the contract, specifications, and drawings. Further production of the arming device by the contractor prior to approval of the contracting activity or the completion of tests on the preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.3 Lot size. As applied to Government inspection of units of product, the term "lot" shall mean "inspection lot," i.e., a collection of units of product submitted by a supplier for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" will be as determined by the Government inspector and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.1.4 Periodic production samples. From each succeeding lot or grand lot of 300 or less, a random sample of 5 arming devices will be selected by the Government inspector for submission to a Bureau of Naval Weapons assigned activity to determine compliance with the requirements of the contract, specifications, and drawings.

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4.2 Contractors' Inspection. Contractors' inspection shall conform to 4.1.

4.3 Classification of tests. The inspection and testing of the arming device shall be classified as follows:

4..3.1 Acceptance tests. These tests, detailed in 4.5, are to be accomplished by the manufacturer on each arming device submitted for acceptance under contract. Inspection and test procedures shall be submitted for Government approval prior to commencement of production. Failure in any test shall be cause for rejecting the arming device.

4.3.2 Preproduction and periodic production tests. preproduction and periodic production tests are those which are accomplished on samples selected as specified in 4.1.2 and 4.1.4 which are representative of the production of the arming device titer the award of the contract to determine that the lot and production meet the requirements of this specification. These tests are detailed in 4.6. Failure of any sample unit in any requirement or test shall result in the rejection of the lot or the cessation of production as determined by the procuring activity.

4.4 Test equipment The following items of test equipment are required to perform the acceptance tests set forth in this specification:

4.4.1 Two dummy actuators having open circuit characteristics.

4.4.2 A suitable torque indicating instrument capable of measuring torques in the order of 8 inch pounds with an accuracy of 5 percent.

4.4.3 Two dummy actuators internally shorted to show a resistance of less than 0.05 ohm between the prongs.

4.4.4 An ohmmeter, with an integral source of 1-1/2 volts direct current (dc) maximum and a reading accuracy of plus or minus 2 percent, capable of measuring resistances in the order of 0.05 ohm.

4.4.5 A megohmmeter with an integral source of 500 volts dc and a midscale reading in the order of 10 megohms.

4.4.6 A spring scale capable of measuring zero to ten pounds.

4.5 Acceptance tests

4.5.1 Test conditions. Unless otherwise specified, the arming device shall be subjected to acceptance tests under the following conditions:

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40.5.1.1 Temperature: Room ambient, plus 65 degrees F to plus 95 degrees F (plus 18.33 degrees C to plus 35.0 degrees C)

4.5.1.2 Altitude: Normal ground

4.5.1.3 Vibration: None

4.5.1.4 Humidity: Room ambient to 95 percent relative (maximum)

4.5.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity, and where definite requirements are not specified, it shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4..5.3 Test procedure. The acceptance tests shall be performed in accordance with the following procedures:

4..5.3.1 Degree of cure of plastic components test. Conduct at least one test for each batch of uncured resin and for each change in the curing procedure. Select a sufficient number of sample upper contact mounts and lower contact mounts to conduct an acetone extraction test on each as prescribed in method No. 7021 of Federal Test Method Standard No. 406. In cases where the extract exceeds 5 percent all contact mounts represented by the sample shall be rejected.

4.5.3.2 Stitch securement test. Attach the spring scale (4.4.6) to the two terminals of the switch and apply a pull of three pounds minimum.

4.5.3.3 Switch operation test. Using the spring scale (4.4.6), "apply a force of four pounds maximum against the switch actuating face and apply the test probes of the ohmmeter (4.4.4) to the terminals of the switch .

4.5.3.4 Operating torque test. Insert two dummy actuators (4.4.1) in the shutter and assemble the shutter and contact mount in the housing without the safety spring. Attach a suitable torque indicating instrument (4.4.2) to the operating shaft of the-shutter and apply a torque, first in the arming direction and then in the disarming direction, until the shutter comes to a full stop.

4.5.3.5 Contact finger test. Insert the internally shorted dummy actuators (4.4.3) in the shutter, assemble the shutter and contact mount in the housing, and rotate the shutter between the contact fingers. Apply the test probes of the ohmmeter (4.4.4) to the terminal pins of the contact mount assembly.

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4.5.3.6 Shunt contact resistance. Attach a short lead wire whose resistance is less than 0.03 ohm to each prong of a dummy actuator (4.4.1) and insert the actuator in one of the recesses in the shutter. Assemble the shutter to the contact mount assembly and rotate it between the shunt fingers. Apply the test probes of an ohmmeter to the leads from the actuator. Repeat this test with the same dummy actuator inserted in the other recess of the shutter.

4.5.3.7 Contact mount assembly. This test is performed on the contact mount assembly, Dwg 883562. Apply the test probes of an ohmmeter (4.4.4) to the two terminal pins.

4.,5.3.8 Dielectric test. On the contact mount assembly apply the test probes of a megohmmeter 4.5) between either terminal pin and the rivets of the shunt fingers. Then assemble the shutter and contact mount assembly in the housing, without actuators and without the safety spring. Measure the insulation resistance between each terminal pin and the shutter shaft, and between each terminal pin and the housing.

4..6 Preproduction and periodic production tests

4.6.1 Vibration. The mechanism shall be tested in two conditions, as described below.

(a) The first condition shall be with the arming device mounted rigidly to the vibration table by means of a mounting method simulating the actual method of mounting used in service. The vertical displacement component of the vibration shall be simple harmonic motion. The samples shall be placed on the vibration table so that equal quantities of devices experience vibration perpendicular to the plane of mounting and vibration parallel to the plane of mounting. The mechanism shall be vibrated through a range of frequencies from 700 to 5400 cycles per minute in 100 cycles per minute increments. The duration of the vibration of each increment and the, amplitude of the vibration shall be in accordance with the following schedule:

FREQUENCY (cycles per minute)	AMPLITUDE (inches)	DURATION
700 to 1500	0.030 ± 0.002	1 hour
1600 to 2000	0.020 ± 0.002	1 hour
2100 to 2500	0.013 ± 0.001	1 hour
2600 to 3000	0.009 ± 0.001	1 hour
3100 to 3500	0.004 ± 0.001	5 minutes
3600 to 4000	0.003 ± 0.0005	5 minutes
4100 to 4500	0.002 ± 0.0005	5 minutes
4600 to 5400	.0015 ± 0.0005	5 minutes

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(b) The second condition shall be with the device in the shipping container as specified in section 5. The mechanisms, packed in the shipping container, shall be mounted on a vibration platform equipped with barriers so as to allow free movement of the container relative to the -platform but prevent the devices from falling off the platform. The motion of the platform shall be such that any point on the platform moves in a circular path one inch in diameter with the plane of the circle perpendicular to the platform. The frequency shall be within the range of 240 to 300 cycles per minute. The package of mechanisms shall be vibrated for a period of one-half hour on each face of the container.

4.6.2 Shock. The arming device under test shall be subjected to a continuous two-phase shock applied in any direction.

(a) The first (impact) phase shall consist of a change of velocity of 17.0 to 19.0 feet per second taking place in from 0.2 to 0.4 millisecond.

(b) During the second (drag) phase, the arming device shall be accelerated for a minimum time of 0.15 millisecond ‘in such a manner that during this time the minimum average acceleration is 125 gravity units.

4..6.3 Temperature. Each arming device shall be capable of properly functioning throughout the temperature range of minus 65 degrees F to plus 160 degrees F.

4.6.4 Storage. As inclosed and sealed within the inner container, Arming Device Mark 6 Mod O (Torpedo Practice) shall withstand without damage a three day storage period at minus 80 degrees F.

4.6.5 Acceptance criteria. Failure of the arming device to meet any of the requirements and tests of this specifications shall be considered cause for rejection of the lot.

5* PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Level A

5 .1.1.1 Each Arming Device Mark 6 Mod 0 (Torpedo Practice) shall be individually packaged in accordance with LD 291935.

5.1.2 Level C

5.1.2.1 Arming Devices shall be individually packaged in a manner which will afford adequate protection from deterioration and physical damage during shipment from the contractor or supply source to the first receiving activity for immediate use.

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5.2 Packing

5.2.1 Level A

5.2.1.1 Twenty-four unit containers shall be packed in accordance with LD 291936.

5.2.2 Level C

5.2.2.1 Arming devices shall be packed in a manner which will provide protection from damage during direct shipment from the contractor or supply source to the first receiving activity for immediate use. The type of containers used shall conform to applicable carrier rules and regulations.

5.3 Marking

5.3.1 In addition to any special marking required by the contract or order, all marking shall be in accordance with Military Standard MIL-STD-129

CAUTION: At the time of packaging Arming Device Mark 6 Mod 0 (Torpedo Practice) and subsequent removal from the package for assembly with actuators, care shall be exercised to avoid losing or damaging either the retaining ring, Dwg 883581, or spring washer, Dwg 959425, as these components are required for assembly of the arming device with the exploder.

6. NOTES

6.1 Ordering data. Procurement documents should specify the desired level of packaging and packing. Level C should not be selected unless it is definitely known that the equipment will be used immediately upon receipt.

NOTICE

When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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